ITCS216: Database Management Systems (1/2026)

Faculty of Information and Communication Technology

Workshop on Project Phase 1: Data Modeling

This workshop aims to provide a chance to refine the business rules and develop the data model for Project Phase I. All members of the team are required to analyze the business requirements (both rules and constraints) and sketch the data model using ERD Chen Notations.

Instructions:

1. **Use the group project to do an individual assignment.**
2. Starting from the M1 report, **each member** picks the different business process.
3. Revise / or create at least 3 business rules using **Template#1.**

Template#1: **Format:** Noun + Verb + Noun (e.g., a student enrolls many classes)

Business Process Name: Enrollment		
Rule No. 1	**Rule Statement**	**Relationship**
Forward/Active	A student can enroll many classes.	1:M
Backward/Passive	A class can be enrolled by many students.	1:M
Conclusion	Students can enroll many classes.	M:N

Note: you need to formulate both Backward and Forward

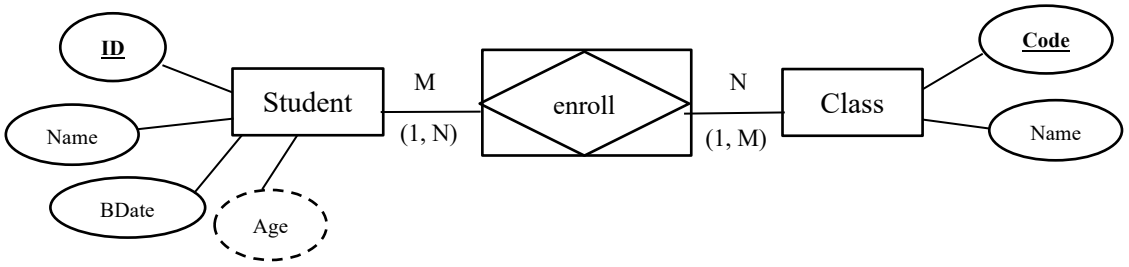
4. Revise / or create at least 2 business rules using **Template#2.**

Template#2: **Format**: Noun + Verb + Attributes (e.g., a student comprises of ID, Name, Email),

Business Process Name: Enrollment
Rule No. 2
A student comprises of ID, Name, Birthdate, and calculated Age.

5. Use the results obtained from Step 2-3 to draw the ERD (CHEN notation) using **Template #3**. All necessary notations for each rule have to demonstrate with complete and correct design

Template #3: Example

Business Process Name: Enrollment	
Rule No. 1	Students can enroll many classes. (M:N)
Rule No. 2	A student comprises of ID, Name, Birthdate, and calculated Age.
Rule No. X	A class consists of class code and name.
...	
ERD 	

6. Use the result from your teammate to create only one EERD diagram per group. (Recommended Tools: https://www.drawio.com/)

Template#1: Format: Noun + Verb + Noun

Business Process Name: Member registration		
Rule No. #	**Rule Statement**	**Relationship**
Forward/Active	A club registers many members	1:M
Backward/Passive	A member is registered at one club	1:1
Conclusion	A club registers many members	1:M

Template#1: Format: Noun + Verb + Noun

Business Process Name: Member registration		
Rule No. #	**Rule Statement**	**Relationship**
Forward/Active	A member accesses many same-tier gyms.	1:M
Backward/Passive	A same-tier gym is accessed by many members.	1:M
Conclusion	Many members can access many same-tier gyms.	M:N

Template#1: Format: Noun + Verb + Noun

Business Process Name: Member registration		
Rule No. #	**Rule Statement**	**Relationship**
Forward/Active	A member selects one membership plan.	1:M
Backward/Passive	A membership plan is selected by many members.	1:M
Conclusion	Many membership plans can be selected by many members.	M:N

Template#2: Format: Noun + Verb + Attributes

Business Process Name: Member registration
Rule No. #
A member comprises of Member_ID, First_Name, Last_Name, ID_Number,Address, Phone_Number, Email_Address, and Timestamp

Template#2: Format: Noun + Verb + Attributes

Business Process Name: Member registration
Rule No. #
A same-tier gym comprises of Gym_ID, Gym_Name, Gym_Tier, and Gym_Address.